EXP 1: To implement basic Linux commands on Ubuntu.

1) Ls command : The ls command in Linux is used to list files and directories present in current working directories.

```
sit@sit-OptiPlex-3280-AIO: ~
sit@sit-OptiPlex-3280-AIO:~$ ls
1078_practical2  1111.l  78_os  a.out  Desktop  Documents  Downloads  lap8.l  lap8.y  lex.yy.c  Music  number_1221.l  Pictures  Public  snap  Templates  test  Videos  y.tab.c  y.tab.h
sit@sit-OptiPlex-3280-AIO:~$ ls -l
total 164
drwxrwxr-x 2 sit sit  4096 Jan 21 09:18 1078_practical2
-rw-rw-r-- 1 sit sit   362 Jul  3  2025 1111.l
drwxrwxr-x 2 sit sit  4096 Jan 21 10:43 78_os
-rwxrwxr-x 1 sit sit 31920 Nov 12 14:14 a.out
drwxr-xr-x 2 sit sit  4096 Nov 12 13:43 Desktop
drwxr-xr-x 2 sit sit  4096 Apr 11  2025 Documents
drwxr-xr-x 2 sit sit  4096 Jan 21 10:36 Downloads
-rw-rw-r-- 1 sit sit   790 Nov 12 14:09 lap8.l
-rw-rw-r-- 1 sit sit  1129 Nov 12 14:14 lap8.y
-rw-rw-r-- 1 sit sit 46569 Nov 12 14:14 lex.yy.c
drwxr-xr-x 2 sit sit  4096 Apr 11  2025 Music
-rw-rw-r-- 1 sit sit   201 Jul  4  2025 number_1221.l
drwxr-xr-x 3 sit sit  4096 Jul  3  2025 Pictures
drwxr-xr-x 2 sit sit  4096 Apr 11  2025 Public
drwx------ 6 sit sit  4096 Nov  6 12:12 snap
drwxr-xr-x 2 sit sit  4096 Apr 11  2025 Templates
-rwxrwxr-x 1 sit sit 15960 Apr 11  2025 test
drwxr-xr-x 2 sit sit  4096 Apr 11  2025 Videos
-rw-rw-r-- 1 sit sit    74 Jan 22 10:42 y.tab.c
-rw-rw-r-- 1 sit sit  3728 Nov 12 14:14 y.tab.h
sit@sit-OptiPlex-3280-AIO:~$ ls -a
.   1078_practical2  78_os  .bash_history  .bashrc  .config  Documents  .gnupg  lap8.l  .lesshst  .local  number_1221.l  .profile  snap  .sudo_as_admin_successful  test    y.tab.c
..  1111.l           a.out  .bash_logout   .cache   Desktop   Downloads  .java   lap8.y  lex.yy.c  Music   Pictures       Public    .ssh  Templates                   Videos  y.tab.h
sit@sit-OptiPlex-3280-AIO:~$ ls -A
1078_practical2  78_os  .bash_history  .bashrc  .config  Documents  .gnupg  lap8.l  .lesshst  .local  number_1221.l  .profile  snap  .sudo_as_admin_successful  test    y.tab.c
1111.l           a.out  .bash_logout   .cache   Desktop   Downloads  .java   lap8.y  lex.yy.c  Music   Pictures       Public    .ssh  Templates                   Videos  y.tab.h
sit@sit-OptiPlex-3280-AIO:~$ ls -d
.
sit@sit-OptiPlex-3280-AIO:~$ ls -i
4592990 1078_practical2  4587866 a.out      4587595 Downloads  4594045 lex.yy.c       4587600 Pictures  4587596 Templates  4594246 y.tab.c
4588313 1111.l           4587594 Desktop    4587741 lap8.l     4587599 Music          4587597 Public    4589005 test       4594235 y.tab.h
4588151 78_os            4587598 Documents  4589037 lap8.y     4588191 number_1221.l  4587529 snap      4587601 Videos
```

2)rmdir command: It is used to remove a directory.

```
sit@sit-OptiPlex-3280-AIO:~/semester4$ rmdir semester4
rmdir: failed to remove 'semester4': No such file or directory
```

3)rm command: It is used to remove or delete files permanently.

```
sit@sit-OptiPlex-3280-AIO:~$ rm kk.txt
sit@sit-OptiPlex-3280-AIO:~$ rm kanak.txt
```

4)uname command: This command is used to display system information.

Whoami command: This command tells who the user is on the system.

```
sit@sit-OptiPlex-3280-AIO:~/semester4$ whoami
sit
sit@sit-OptiPlex-3280-AIO:~/semester4$ uname
Linux
sit@sit-OptiPlex-3280-AIO:~/semester4$
```

5)man command: The man command is used to display command manuals.

```
sit@sit-OptiPlex-3280-AIO:~$ man ls
sit@sit-OptiPlex-3280-AIO:~$ man cp
sit@sit-OptiPlex-3280-AIO:~$
sit@sit-OptiPlex-3280-AIO:~$ man cd
No manual entry for cd
```

6)touch command : It is used to create empty files.

```
sit@sit-OptiPlex-3280-AIO:~$ touch kanak.txt
sit@sit-OptiPlex-3280-AIO:~$ touch kk.txt
```

7) mv command : It is used to move or rename files and directories.

```
sit@sit-OptiPlex-3280-AIO:~/semester4$ touch food.txt lunch.txt
sit@sit-OptiPlex-3280-AIO:~/semester4$ mv food.txt lunch.txt
sit@sit-OptiPlex-3280-AIO:~/semester4$
```

8)cp command : It is used to copy files or directories.

```
sit@sit-OptiPlex-3280-AIO:~$ cp kanak.txt kk.txt
sit@sit-OptiPlex-3280-AIO:~$ cp y.tab.c kanak.txt
```

9)mkdir command : It is used to create new directory.

```
sit@sit-OptiPlex-3280-AIO:~$ mkdir semester4
sit@sit-OptiPlex-3280-AIO:~$
```

10) pwd command : This command displays the content that is currently running in directory.

```
sit@sit-OptiPlex-3280-AIO:~/semester4$ pwd
/home/sit/semester4
```

11) cd command : cd command is used to change current working directory.

```
sit@sit-OptiPlex-3280-AIO:~$ cd semseter4
bash: cd: semseter4: No such file or directory
sit@sit-OptiPlex-3280-AIO:~$ cd semester4
sit@sit-OptiPlex-3280-AIO:~/semester4$
```

12) less command : The less command is a powerful command-line utility used to view the contents of a file or command output one screen at a time.

```
sit@sit-OptiPlex-3280-AIO:~/semester4$ less kt.txt
sit@sit-OptiPlex-3280-AIO:~/semester4$
```

13) echo command : It is used to print the required string.

```
sit@sit-OptiPlex-3280-AIO:~/semester4$ MYPRN=290
sit@sit-OptiPlex-3280-AIO:~/semester4$ echo MYPRN
MYPRN
sit@sit-OptiPlex-3280-AIO:~/semester4$ echo "I am doing Linux Commands"
I am doing Linux Commands
sit@sit-OptiPlex-3280-AIO:~/semester4$ echo "I am studying Operating System"
I am studying Operating System
sit@sit-OptiPlex-3280-AIO:~/semester4$
```